Voot Tangkaratt

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Profile summary

I am a researcher and engineer specializing in **imitation, reinforcement, and reward learning**. My goal is to develop physical AI systems that work reliably in the real-world. I have been based in Japan since 2012, working across academic and industrial R&D environments to develop and deploy deep learning models onto physical systems.

Work history

Sony Research/Sony AI, Japan

Senior Research Scientist | 2025 - *Present*

- Managed a small team of researchers and engineers on daily technical execution.
- Collaborated directly with industrial domain experts to extract practical knowledge and translate it into robot learning systems.

Research Scientist | 2022 - 2025

- Researched and developed robot learning pipelines and algorithms for fine manipulation tasks in gastronomy robotics.
- Built and maintained a major portion of the team's large-scale Python codebase.

RIKEN Advanced Intelligence Project, Japan

Postdoctoral Researcher | 2017 – 2021

- Researched and developed robust imitation and reinforcement learning algorithms under real-world data constraints.
- Published research findings at international AI/ML conferences.
- Collaborated with external research and industrial organizations for publications and patents.

Technical skills

Research & Engineering:

- **Research, develop, and fine-tune** robot policy networks (including Vision-Language-Action (VLA) architectures) utilizing imitation, reinforcement, and reward learning frameworks in simulation and on real-world hardware.
- **Implement and optimize** deep learning models and algorithms using PyTorch, Jax, Keras, and TensorFlow.
- **Design and implement** pipelines to collect multi-modal data from robot manipulators using teleoperation, kinesthetic teaching, motion capture, RGB-D cameras, and tactile sensors.
- **Deploy** trained learning models to control real-world robot manipulators.

Programming & Tools:

- **Python** (Core language; own and maintain a major portion of large scale codebases)
- **C++** (Familiar language; analyze and review logic; personal prototyping for self-study)
- **Robotics & Simulation**: ROS ecosystem, MoveIt, Impedance controllers, RGB-D and tactile data processing, and MuJoCo.
- **Software:** AWS GPU clusters, Git, CI/CD workflows, and Docker.

Education

- **PhD in Computer Science**

  The University of Tokyo, Japan | 2017

  Advisor: Prof. Masashi Sugiyama (Machine Learning laboratory)

- **Masters in Computer Science**

  Tokyo Institute of Technology, Japan | 2014

  Advisor: Prof. Masashi Sugiyama (Machine Learning laboratory)

- **Bachelors in Computer Engineering**

  Chulalongkorn University, Thailand | 2011

Language

- **English:** Academic and business level
- **Japanese:** Intermediate level
- **Thai:** Native speaker

Selected publications

- **Voot Tangkaratt,** Nontawat Charoenphakdee, Masashi Sugiyama. “Robust Imitation Learning from Noisy Demonstrations”. AISTATS, 2021.
- **Voot Tangkaratt**, Bo Han, Mohammad Emtiyaz Khan, Masashi Sugiyama. “Variational Imitation Learning with Diverse-quality Demonstrations”. ICML, 2020.
- Tatsuya Tanaka, Toshimitsu Kaneko, Masahiro Sekine, **Voot Tangkaratt**, Masashi Sugiyama. “Simultaneous Planning for Item Picking and Placing by Deep Reinforcement Learning”. IROS, 2020.
- Yueh-Hua Wu, Nontawat Charoenphakdee, Han Bao, **Voot Tangkaratt**, Masashi Sugiyama. “Imitation Learning from Imperfect Demonstration”. ICML, 2019.
- **Voot Tangkaratt,** Abbas Abdolmaleki, Masashi Sugiyama. “Guide actor-critic for continuous control”. ICLR, 2018.
- Mohammad Khan, Didrik Nielsen, **Voot Tangkaratt**, Wu Lin, Yarin Gal, Akash Srivastava. “Fast and Scalable Bayesian Deep Learning by Weight-perturbation in Adam”. ICML, 2018.

(Full publication list and academic repositories are available on Google Scholar and GitHub. Post-2022 industrial R&D codebases are proprietary assets.)